Week 02 teacher guide

Ratios and matching amounts

Keep this guide beside PowerPoint slides 36-72. Follow the numbered slide sequence, then use the matching answer section after the student has tried.

The seven days

Day 008. Saturday | pages 36-42 | slides 36-53
Equivalent ratios

Day 009. Sunday | pages 43-52 | slides 54-72
Ratio tables and comparisons

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Ratios and matching amounts

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Ratios and matching amounts

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Ratios and matching amounts

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Weekly check

This week’s idea

A ratio compares two amounts in a chosen order. Equivalent ratios scale both amounts together. Tables and double number lines keep matching amounts connected.

Use the built-in Booster review where it appears in the day’s work. It replaces ordinary practice. No separate Booster packet is needed.

PDF bookmarks jump to each lesson and answer section. The student workbook’s Sunday reference pages are for checking an example after an attempt. Keep this teacher guide and the answer-review slides out of view during independent work.

Saturday. Equivalent ratios

Day 008, pages 36-42 | PowerPoint slides 36-53

Goal: I can keep a ratio unchanged while making more or less of a mixture.

Set out: two colors of counters, paper strips, pencil.

Check first: Multiplication, division, and equal groups.

Pacing: Retrieval 7; launch 8; explore 20; discuss 15; synthesis 7; exit 8; preview 5; slack 5. Total 75 min.

Explore: what to watch for

Says four red for every three blue. Why it matters: Names both quantities and their order. Tutor move: Ask for the reversed ratio and compare meanings.

Repeats or partitions a complete group. Why it matters: Shows equivalent ratios concretely. Tutor move: Ask which factor acts on both amounts.

Uses a half-batch or a unit batch. Why it matters: Non-whole factors can preserve the ratio. Tutor move: Accept either method if both quantities scale equally.

Likely misconceptions

Adds the same count to both colors. Why: Confuses additive and multiplicative change. Tutor move: Build identical batches.

Says 4 : 3 means four out of three. Why: Confuses part-to-part and part-to-whole. Tutor move: Count the total seven tiles before naming the red fraction.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 36. Week 2, Saturday

Topic: Equivalent ratios
Student: Day 008, pages 36-42

Say: I can keep a ratio unchanged while making more or less of a mixture.

Ask: What do you already know about this idea?

Expect: Multiplication, division, and equal groups.

If stuck: Ask for the reversed ratio and compare meanings.

Slide 37. Today

Topic: Equivalent ratios
Student: Day 008, pages 36-42

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: Equivalent ratios multiply or divide both quantities by the same factor.

If stuck: Read the learning goal together.

Slide 38. Recall | 4 min

Topic: Equivalent ratios

Student: Day 008, pages 36-42

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: $\frac{1}{6}$ liter; 2: $\frac{5}{8} > \frac{3}{5}$; 3: $\frac{1}{3} > \frac{1}{4}$; 4: $\frac{3}{5}$

If stuck: Give time to draw before giving a hint.

Slide 39. Recall: answers | 3 min

Topic: Equivalent ratios

Student: Day 008, pages 36-42

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: $\frac{1}{6}$ liter; $\frac{5}{8} > \frac{3}{5}$; $\frac{1}{3} > \frac{1}{4}$; $\frac{3}{5}$

If stuck: Rebuild one missed item with its model.

Slide 40. Launch | 8 min

Topic: Equivalent ratios

Student: Day 008, pages 36-42

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: Adding the same number to both counts keeps a mixture the same.

Expect: P1: 4 : 3; 3 : 4

If stuck: Ask for the reversed ratio and compare meanings.

Slide 41. Explore | 4 min

Topic: Equivalent ratios

Student: Day 008, pages 36-42

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: Adding the same number to both counts keeps a mixture the same.

Expect: P2: 9 nuts; P3: 4 : 3

If stuck: Ask for the reversed ratio and compare meanings.

Slide 42. The model | 4 min

Topic: Equivalent ratios

Student: Day 008, pages 36-42

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: Repeat the whole 4 : 3 group.

If stuck: Ask for the reversed ratio and compare meanings.

Slide 43. Required practice | 3 min

Topic: Equivalent ratios
Student: Day 008, pages 36-42

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: Adding the same number to both counts keeps a mixture the same.

Expect: P4: 6 cups; P5: 6 cups yellow

If stuck: Ask for the reversed ratio and compare meanings.

Slide 44. Required practice | 3 min

Topic: Equivalent ratios
Student: Day 008, pages 36-42

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: Adding the same number to both counts keeps a mixture the same.

Expect: P6: 12 builders

If stuck: Ask for the reversed ratio and compare meanings.

Slide 45. Watch the thinking | 3 min

Topic: Equivalent ratios
Student: Day 008, pages 36-42

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 8 cups seeds

If stuck: Return to the matching unit in the model.

Slide 46. Your turn | 3 min

Topic: Equivalent ratios
Student: Day 008, pages 36-42

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 2 batches; 4 cups seeds

If stuck: Compare with the previous example without copying its numbers.

Slide 47. Compare methods | 8 min

Topic: Equivalent ratios
Student: Day 008, pages 36-42

Say: Partner compare: show the same problem in two ways.

Ask: Agree or disagree: Adding the same number to both counts keeps a mixture the same. Partner compare: How do you know the recipe with nine cups of oats has the same taste?

Expect: Accept drawings and equations that preserve the same quantities and units. Repeats or partitions a complete group.

If stuck: Ask which factor acts on both amounts.

Slide 48. A claim to check | 7 min

Topic: Equivalent ratios
Student: Day 008, pages 36-42

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: No. Two copies would give 8 : 6, not 7 : 6.

If stuck: Build two full copies of the original group and compare.

Slide 49. Words and meaning | 4 min

Topic: Equivalent ratios
Student: Day 008, pages 36-42

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: 4 red to 3 blue is 4 : 3. 4 : 3 and 8 : 6 are equivalent. From 4 : 3 to 12 : 9, the factor is 3.

If stuck: Read one definition and draw its example.

Slide 50. In your own words | 3 min

Topic: Equivalent ratios
Student: Day 008, pages 36-42

Say: Use your own words and one example.

Ask: Agree or disagree: Adding the same number to both counts keeps a mixture the same.

Expect: Equivalent ratios multiply or divide both quantities by the same factor.

If stuck: Point to the model and name what each part represents.

Slide 51. Exit ticket | 8 min

Topic: Equivalent ratios
Student: Day 008, pages 36-42

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: 21 green beads

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 52. The week ahead | 5 min

Topic: Equivalent ratios
Student: Day 008, pages 36-42

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 53. Tutor reference

Topic: Equivalent ratios

Student: Day 008, pages 36-42

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: 21 green beads

Got it: Gives 21 green beads and shows a factor of 3 on both quantities.

Almost there: Gives 21 with incomplete labeling, or makes a small arithmetic slip after using factor 3.

Needs re-teaching: Gives 15 or another answer from adding the same amount.

Sunday. Ratio tables and comparisons

Day 009, pages 43-52 | PowerPoint slides 54-72

Goal: I can use tables, double number lines, and equal totals to solve ratio problems.

Set out: grid paper, ruler, two colors of counters.

Check first: Scaling both amounts in a ratio.

Pacing: Retrieval 5; launch 5; explore 15; discuss 12; synthesis 5; mixed review 10; exit 8; preview 5. Total 65 min.

Explore: what to watch for

Keeps matching values in the same row. Why it matters: Pairs are meaningful together. Tutor move: Ask what one complete row means.

Finds a common water amount or compares part fractions. Why it matters: Both are valid ratio comparisons. Tutor move: Accept either method and require a shared comparison unit.

Adds ratio parts before using the total. Why it matters: Separates a part-to-part ratio from the whole. Tutor move: Ask the student to label all five parts.

Likely misconceptions

Compares mixtures by subtraction. Why: A difference ignores mixture size. Tutor move: Use the same amount of water in both mixtures.

Divides total by one ratio number. Why: Treats one part as the whole. Tutor move: Draw and count all ratio parts first.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 54. Week 2, Sunday

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: I can use tables, double number lines, and equal totals to solve ratio problems.

Ask: What do you already know about this idea?

Expect: Scaling both amounts in a ratio.

If stuck: Ask what one complete row means.

Slide 55. Today

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: A ratio table records equal groups. Comparing mixtures needs a common amount or a common total.

If stuck: Read the learning goal together.

Slide 56. Recall | 3 min

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: At 1 $\frac{3}{4}$, the seventh quarter-step from 0; 2: $\frac{5}{12}$; 3: Smaller; 6; 4: 2

If stuck: Give time to draw before giving a hint.

Slide 57. Recall: answers | 2 min

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: At 1 $\frac{3}{4}$, the seventh quarter-step from 0; $\frac{5}{12}$; Smaller; 6; 2

If stuck: Rebuild one missed item with its model.

Slide 58. Launch | 5 min

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: A mixture with more cups of juice always tastes stronger.

Expect: P1: 10 cups water

If stuck: Ask what one complete row means.

Slide 59. Explore | 3 min

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: A mixture with more cups of juice always tastes stronger.

Expect: P2: 25 cups water

If stuck: Ask what one complete row means.

Slide 60. The model | 3 min

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: Two cups concentrate pairs with five cups water.

If stuck: Ask what one complete row means.

Slide 61. Required practice | 3 min

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: A mixture with more cups of juice always tastes stronger.

Expect: P3: Mix A; P4: 14 cups yellow and 21 cups blue

If stuck: Ask what one complete row means.

Slide 62. Required practice | 2 min

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: Try these before I show a method. Use your matching student workbook pages.

Ask: Agree or disagree: A mixture with more cups of juice always tastes stronger.

Expect: P5: Yes

If stuck: Ask what one complete row means.

Slide 63. Watch the thinking | 2 min

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 15 orange, 10 purple

If stuck: Return to the matching unit in the model.

Slide 64. Your turn | 2 min

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 6 per part; 24 green; 6 white

If stuck: Compare with the previous example without copying its numbers.

Slide 65. Compare methods | 6 min

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: Partner compare: show the same problem in two ways.

Ask: Agree or disagree: A mixture with more cups of juice always tastes stronger. Partner compare: Why must the 35-cup paint total be divided into five parts?

Expect: Accept drawings and equations that preserve the same quantities and units. Finds a common water amount or compares part fractions.

If stuck: Accept either method and require a shared comparison unit.

Slide 66. A claim to check | 6 min

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: There are five parts in all. Yellow is $\frac{2}{5}$ of the paint.

If stuck: Join the two yellow parts and three blue parts into one total strip.

Slide 67. Words and meaning | 3 min

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: (2, 5), (4, 10), and (6, 15). Two cups concentrate matches five cups water. 2 red and 3 blue makes red $\frac{2}{5}$ of all tiles.

If stuck: Read one definition and draw its example.

Slide 68. Mixed review | 10 min

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: Choose a method for each problem.

Ask: What tells you which operation to use?

Expect: Larger; 8; 6 pieces; $\frac{7}{8}$

If stuck: Name the quantities and units before calculating.

Slide 69. In your own words | 2 min

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: Use your own words and one example.

Ask: Agree or disagree: A mixture with more cups of juice always tastes stronger.

Expect: A ratio table records equal groups. Comparing mixtures needs a common amount or a common total.

If stuck: Point to the model and name what each part represents.

Slide 70. Exit ticket | 8 min

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: Mix A

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 71. The week ahead | 5 min

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 72. Tutor reference

Topic: Ratio tables and comparisons
Student: Day 009, pages 43-52

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: Mix A

Got it: Chooses A and compares equivalent ratios or correct fractions of juice.

Almost there: Chooses A but does not show a valid comparison, or makes a minor arithmetic slip with the right setup.

Needs re-teaching: Says equal because the differences match, or compares only the juice counts.

Weekday teaching and review

Use one day at a time. Aim for 15-20 minutes, or 15 minutes during a holiday week. Accuracy and a clear explanation matter more than finishing quickly.

Day 010. Monday | pages 53-54

Ask for a model from Saturday before checking an answer. Let the student choose a method. After checking, have the student correct the first step that changed the quantity.

Built-in Booster review: Ratios and rates: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 011. Tuesday | pages 55-56

Connect the practice to Sunday's model. Ask what each number counts and how a multiplication check, diagram, or estimate supports the answer.

Built-in Booster review: Ratios and rates: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 012. Wednesday | pages 57-58

Let the student choose the method for each mixed problem. For the puzzle, ask which conditions are already met. Accept any solution that meets every condition.

Built-in Booster review: Weekly puzzle: items 13. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 013. Thursday | pages 59-60

Ask the student to explain the planted error before repairing it. Optional Stretch comes after the required work and may be skipped without delaying the next day.

Built-in Booster review: Fractions: compare and combine: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 014. Friday | pages 61-62

Give the weekly check independently, without coaching or a calculator. Score the required reasoning as well as the answer. Afterward, choose one error to discuss calmly.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

After Friday

On a 10-item check: 9-10 correct is strong; 7-8 is on track; 5-6 needs another review of the missed topic; 0-4 calls for 20 minutes of re-teaching at the next session. Record the skill and one observation. Check it again with a fresh item at least a week later.

Complete answers for the week

Match the Day number, item ID, and student page before checking. Questions stay in the student workbook; these keys give every answer and the needed reasoning. Equivalent correct forms and sound alternative methods are welcome. An explanation item needs a reason, model, or valid check as well as its conclusion.

Equivalent ratios

Day 008, student workbook pages 36-40

R1 [F-DIV] | student p. 36

Answer: $\frac{1}{6}$ liter

Reasoning: Split the given fractional amount into the stated number of equal shares.

Watch for $\frac{2}{3}$: Multiplying by the divisor answers a different question.

R2 [F-COMP] | student p. 36

Answer: $\frac{5}{8} > \frac{3}{5}$

Reasoning: Rename as twenty-five fortieths and twenty-four fortieths.

Watch for $\frac{5}{8} < \frac{3}{5}$: Compare equal-sized parts, not the original denominator counts.

R3 [F-COMP] | student p. 36

Answer: $\frac{1}{3} > \frac{1}{4}$

Reasoning: With the same numerator, the fraction with larger pieces is larger.

Watch for $\frac{1}{3} < \frac{1}{4}$: A larger denominator makes smaller equal parts, not a larger fraction.

R4 [F-MULT] | student p. 36

Answer: $\frac{3}{5}$

Reasoning: An area model shows 12 of 20 equal parts, or three fifths.

S [EXPLAIN] | student p. 36

Answer: Answers vary. Equivalent ratios multiply or divide both quantities by the same factor.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [R-RATIO] | student p. 37

Answer: 4 : 3; 3 : 4

Reasoning: Name the requested first color before writing its count.

P2 [R-EQUIV] | student p. 37

Answer: 9 nuts

Reasoning: Twelve is three times four, so use three times three nuts.

P3 [R-EQUIV] | student p. 37

Answer: 4 : 3

Reasoning: Make five equal groups. Each has four red and three blue.

P4 [R-EQUIV] | student p. 38

Answer: 6 cups

Reasoning: Nine is 1 $\frac{1}{2}$ times six. Multiply four by the same factor.

P5 [R-EQUIV] | student p. 38

Answer: 6 cups yellow

Reasoning: Fifteen cups blue makes three batches. Each batch uses two cups yellow.

P6 [R-EQUIV] | student p. 38

Answer: 12 builders

Reasoning: Eight testers make four groups of two. Four groups need twelve builders.

F [R-EQUIV] | student p. 39

Answer: 2 batches; 4 cups seeds

Reasoning: Use the worked example in the slides to check each step.

D [R-EQUIV] | student p. 39

Answer: No. Two copies would give 8 : 6, not 7 : 6.

Reasoning: She uses equal additions rather than equal multiplication. Build two full copies of the original group and compare.

E [R-EQUIV] | student p. 40

Answer: 21 green beads

Reasoning: Twelve white beads are three groups of four. Use three groups of seven green.

Watch for 15: Adding eight to each count changes the relationship.

Ratio tables and comparisons

Day 009, student workbook pages 43-48

R1 [F-LINE] | student p. 43

Answer: At 1 $\frac{3}{4}$, the seventh quarter-step from 0

Reasoning: Four fourths reach 1. Three more fourths reach one and three fourths.

R2 [F-SUB] | student p. 43

Answer: $\frac{5}{12}$

Reasoning: Rename both fractions with a common denominator. Subtract equal parts.

R3 [F-SCALE] | student p. 43

Answer: Smaller; 6

Reasoning: Two thirds takes less than all nine.

R4 [F-MULT] | student p. 43

Answer: 2

Reasoning: Count two thirds for each whole-number group, then rename the result.

S [EXPLAIN] | student p. 43

Answer: Answers vary. A ratio table records equal groups. Comparing mixtures needs a common amount or a common total.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [R-TABLE] | student p. 44

Answer: 10 cups water

Reasoning: Double both numbers in (2, 5).

P2 [R-TABLE] | student p. 44

Answer: 25 cups water

Reasoning: Table: multiply (2, 5) by five. Line: five equal jumps of 2 match five jumps of 5.

P3 [R-COMP] | student p. 44

Answer: Mix A

Reasoning: Double A to get 6 syrup : 8 water. Compare 6 syrup with 5 syrup.

P4 [R-WHOLE] | student p. 45

Answer: 14 cups yellow and 21 cups blue

Reasoning: Five equal ratio parts total 35. Each part is 7 cups.

P5 [R-GRAPH] | student p. 45

Answer: Yes

Reasoning: Each y-value is three times its x-value. Zero bags have zero marbles.

F [R-COMP] | student p. 45

Answer: 6 per part; 24 green; 6 white

Reasoning: Use the worked example in the slides to check each step.

D [R-COMP] | student p. 46

Answer: There are five parts in all. Yellow is $\frac{2}{5}$ of the paint.

Reasoning: He uses the other part as the whole. Join the two yellow parts and three blue parts into one total strip.

M1 [F-SCALE] | student p. 47

Answer: Larger; 8

Reasoning: Four thirds takes one whole six plus one third of six.

M2 [F-DIV] | student p. 47

Answer: 6 pieces

Reasoning: Each meter contains 3 equal pieces. Multiply by 2 meters.

Watch for $\frac{2}{3}$: Multiplying by the divisor answers a different question.

M3 [F-COMP] | student p. 47

Answer: $\frac{7}{8}$

Reasoning: The gaps from 1 are one eighth and one fifth. One eighth is smaller.

E [R-COMP] | student p. 48

Answer: Mix A

Reasoning: A is $\frac{4}{7}$ juice; B is $\frac{6}{11}$. Compare $\frac{44}{77}$ and $\frac{42}{77}$.

Watch for They are the same.: Both differences are one, but the fractions of juice differ.

Study guide: Try it

Day 009, student workbook pages 49-52

1 [R-RATIO] | student p. 52

Answer: 5 : 2; 2 : 5

Reasoning: Keep the requested first color first.

2 [R-EQUIV] | student p. 52

Answer: Yes

Reasoning: Multiply both numbers in 6 : 10 by one and one half.

3 [R-TABLE] | student p. 52

Answer: 21

Reasoning: Multiply both entries of the first pair by three.

4 [R-TABLE] | student p. 52

Answer: 32 laps

Reasoning: Four jumps of three minutes match four jumps of eight laps.

5 [R-GRAPH] | student p. 52

Answer: Yes

Reasoning: Every vertical value is twice the horizontal value.

Daily practice

Day 010, student workbook pages 53-54

1 [N-FACT] | student p. 53

Answer: 12

2 [N-FACT] | student p. 53

Answer: 28

3 [N-FACT] | student p. 53

Answer: 16

4 [N-FACT] | student p. 53

Answer: 36

5 [R-EQUIV] | student p. 53

Answer: 3 cups

Reasoning: Find the number of recipe batches by dividing the rice amount by 3. Each batch needs 2 cups beans.

6 [R-EQUIV] | student p. 53

Answer: 20 blue tiles

Reasoning: The number of batches is 12 ÷ 3. Multiply that count by 5.

Watch for 14 blue tiles: Adding the same amount to both counts changes the ratio.

7 [R-EQUIV] | student p. 53

Answer: 10 blue tiles

Reasoning: The number of batches is 6 ÷ 3. Multiply that count by 5.

Watch for 8 blue tiles: Adding the same amount to both counts changes the ratio.

8 [R-EQUIV] | student p. 53

Answer: 15 blue tiles

Reasoning: The number of batches is 9 ÷ 3. Multiply that count by 5.

Watch for 11 blue tiles: Adding the same amount to both counts changes the ratio.

9 [R-RATIO] | student p. 54 | Built-in review: Ratios and rates

Answer: 5 : 2; 2 : 5

Reasoning: Keep the requested first color first.

10 [R-EQUIV] | student p. 54 | Built-in review: Ratios and rates

Answer: Yes

Reasoning: Multiply both numbers in 6 : 10 by one and one half.

11 [R-EQUIV] | student p. 54 | Built-in review: Ratios and rates

Answer: No

Reasoning: Adding two to each count changes the relationship. Two batches would give 6 : 14.

Watch for Yes: Equal additions do not preserve a ratio.

12 [R-EQUIV] | student p. 54 | Built-in review: Ratios and rates

Answer: 24 runners

Reasoning: Eighteen walkers make six groups. Six groups need twenty-four runners.

13 [R-EQUIV] | student p. 54

Answer: No. Two copies would give 8 : 6, not 7 : 6.

Watch for 7 : 6 is equivalent to 4 : 3.: She uses equal additions rather than equal multiplication.

Daily practice

Day 011, student workbook pages 55-56

1 [F-EQUIV] | student p. 55

Answer: 0.5; 50%

2 [F-EQUIV] | student p. 55

Answer: 0.25; 25%

3 [F-EQUIV] | student p. 55

Answer: 0.75; 75%

4 [F-EQUIV] | student p. 55

Answer: 0.125; 12.5%

5 [R-WHOLE] | student p. 55

Answer: 8 yellow; 20 purple

Reasoning: Add ratio parts: 2 + 5 = 7. Divide the total by 7, then multiply each ratio part.

6 [R-TABLE] | student p. 55

Answer: 20

Reasoning: Multiply both entries in the first pair by the same factor.

7 [F-COMP] | student p. 55

Answer: $\frac{1}{3} > \frac{1}{4}$

Reasoning: With the same numerator, the fraction with larger pieces is larger.

Watch for $\frac{1}{3} < \frac{1}{4}$: A larger denominator makes smaller equal parts, not a larger fraction.

8 [N-SCALE] | student p. 55

Answer: 3

9 [R-TABLE] | student p. 56 | Built-in review: Ratios and rates

Answer: 21

Reasoning: Multiply both entries of the first pair by three.

10 [R-TABLE] | student p. 56 | Built-in review: Ratios and rates

Answer: 32 laps

Reasoning: Four jumps of three minutes match four jumps of eight laps.

11 [R-GRAPH] | student p. 56 | Built-in review: Ratios and rates

Answer: Yes

Reasoning: Every vertical value is twice the horizontal value.

12 [R-WHOLE] | student p. 56 | Built-in review: Ratios and rates

Answer: $\frac{3}{8}$

Reasoning: Count all eight cups before forming the fraction.

Watch for $\frac{3}{5}$: Five counts the other part, not the whole.

13 [R-COMP] | student p. 56

Answer: There are five parts in all. Yellow is $\frac{2}{5}$ of the paint.

Watch for $\frac{2}{3}$: He uses the other part as the whole.

Daily practice

Day 012, student workbook pages 57-58

1 [N-SCALE] | student p. 57

Answer: 42

2 [N-SCALE] | student p. 57

Answer: 4

3 [N-SCALE] | student p. 57

Answer: 9

4 [N-SCALE] | student p. 57

Answer: 12

5 [R-EQUIV] | student p. 57

Answer: Yes

Reasoning: Multiply both numbers in 6 : 10 by one and one half.

6 [R-EQUIV] | student p. 57

Answer: No

Reasoning: Adding two to each count changes the relationship. Two batches would give 6 : 14.

Watch for Yes: Equal additions do not preserve a ratio.

7 [F-MULT] | student p. 57

Answer: 2

Reasoning: Count two thirds for each whole-number group, then rename the result.

8 [F-ADD] | student p. 57

Answer: $\frac{5}{6}$

Reasoning: Rename both fractions with a common denominator. Add the counts of equal parts.

9 [D-OPS] | student p. 58

Answer: 4.75

10 [G-VOL] | student p. 58

Answer: 36 cm³

11 [N-FACT] | student p. 58

Answer: 14

12 [G-VOL] | student p. 58

Answer: 42 cm³

13 [PUZZLE] | student p. 58 | Built-in review: Weekly puzzle

Answer: 15 red and 10 blue

Reasoning: One batch contains five tiles. The only multiple of five strictly between 20 and 30 is 25.

Daily practice

Day 013, student workbook pages 59-60

1 [N-EST] | student p. 59

Answer: 400

2 [N-EST] | student p. 59

Answer: 5

3 [N-EST] | student p. 59

Answer: About 100

4 [N-EST] | student p. 59

Answer: About 20

5 [R-EQUIV] | student p. 59

Answer: Yes

Reasoning: Multiply both numbers in 6 : 10 by one and one half.

6 [R-EQUIV] | student p. 59

Answer: No

Reasoning: Adding two to each count changes the relationship. Two batches would give 6 : 14.

Watch for Yes: Equal additions do not preserve a ratio.

7 [R-EQUIV] | student p. 59

Answer: 4 : 7

Reasoning: Divide both counts by the same factor.

8 [F-DIV] | student p. 59

Answer: $\frac{1}{18}$; stories vary

Reasoning: Example: share one sixth liter equally among three cups. Each gets one eighteenth liter.

9 [F-ADD] | student p. 60 | Built-in review: Fractions: compare and combine

Answer: $\frac{5}{6}$

Reasoning: Rename both fractions with a common denominator. Add the counts of equal parts.

10 [F-ADD] | student p. 60 | Built-in review: Fractions: compare and combine

Answer: $\frac{11}{12}$

Reasoning: Rename both fractions with a common denominator. Add the counts of equal parts.

11 [F-ADD] | student p. 60 | Built-in review: Fractions: compare and combine

Answer: $\frac{7}{12}$

12 [F-ADD] | student p. 60 | Built-in review: Fractions: compare and combine

Answer: $\frac{3}{4}$

13 [R-EQUIV] | student p. 60

Answer: 24 runners

Reasoning: Eighteen walkers make six groups. Six groups need twenty-four runners.

14 [R-COMP] | student p. 60

Answer: There are five parts in all. Yellow is $\frac{2}{5}$ of the paint.

Watch for $\frac{2}{3}$: He uses the other part as the whole.

Weekly check

Day 014, student workbook pages 61-62

1 [R-EQUIV] | student p. 61

Answer: Yes

Reasoning: Multiply both numbers in 6 : 10 by one and one half.

2 [R-EQUIV] | student p. 61

Answer: No

Reasoning: Adding two to each count changes the relationship. Two batches would give 6 : 14.

Watch for Yes: Equal additions do not preserve a ratio.

3 [R-EQUIV] | student p. 61

Answer: 4 : 7

Reasoning: Divide both counts by the same factor.

4 [R-EQUIV] | student p. 61

Answer: 4 : 7

Reasoning: Divide both counts by the same factor.

5 [F-COMP] | student p. 61

Answer: $\frac{2}{4} > \frac{2}{5}$

Reasoning: With the same numerator, the fraction with larger pieces is larger.

Watch for $\frac{2}{4} < \frac{2}{5}$: A larger denominator makes smaller equal parts, not a larger fraction.

6 [F-LINE] | student p. 62

Answer: At 1 $\frac{3}{4}$, the seventh quarter-step from 0

Reasoning: Four fourths reach 1. Three more fourths reach one and three fourths.

7 [F-MULT] | student p. 62

Answer: $\frac{3}{5}$

Reasoning: An area model shows 12 of 20 equal parts, or three fifths.

8 [D-OPS] | student p. 62

Answer: 6.75

9 [G-VOL] | student p. 62

Answer: 90 cm^3

10 [R-EQUIV] | student p. 62

Answer: No. Two copies would give 8 : 6, not 7 : 6.

Watch for 7 : 6 is equivalent to 4 : 3.: She uses equal additions rather than equal multiplication.